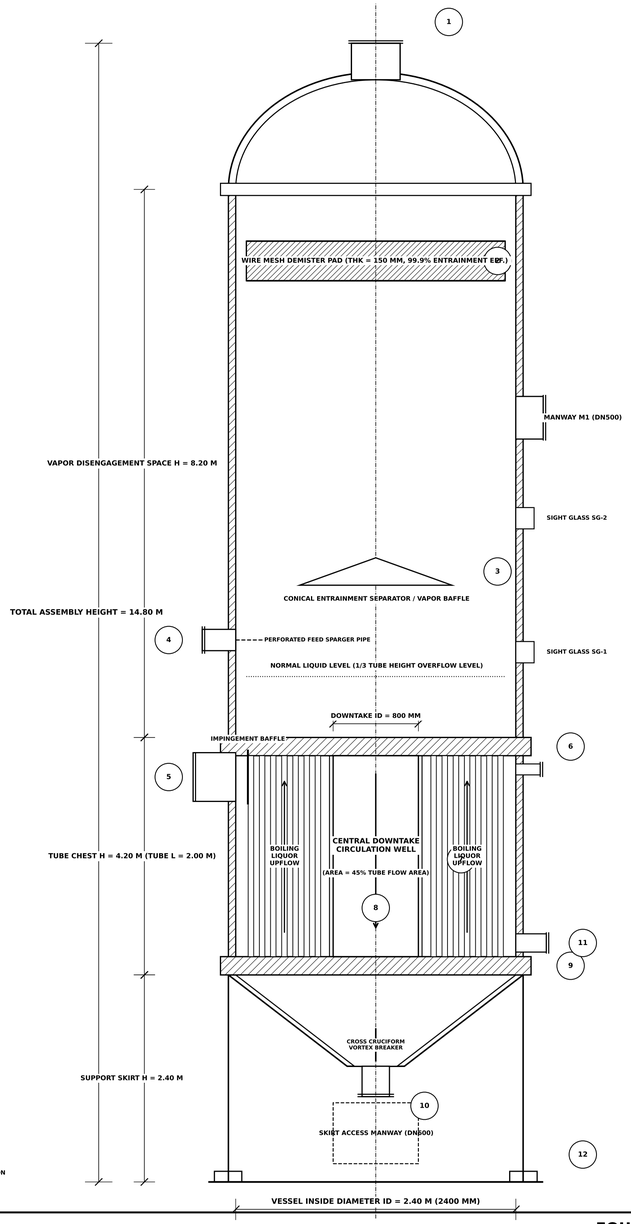
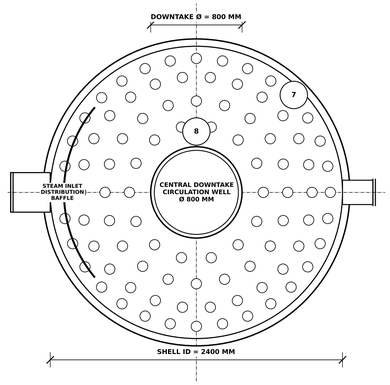
VIEW 1 — DETAILED SECTIONAL ELEVATION (CUT-OPEN VIEW)

SHORT-TUBE VERTICAL CALANDRIA EVAPORATOR (E-101 / E-102 / E-103)



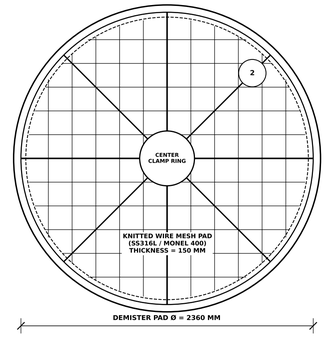
VIEW 2 — SECTION A-A (CALANDRIA TUBE SHEET LAYOUT)

TRIANGULAR TUBE PITCH WITH CENTRAL CIRCULATION DOWNTAKE



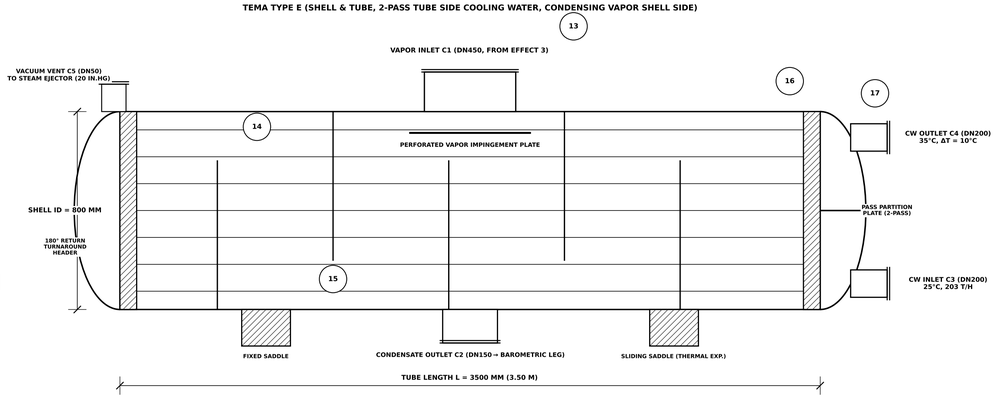
VIEW 3 — SECTION B-B (DEMISTER PAD & HOLD-DOWN GRID)

KNITTED WIRE MESH PAD ON RADIAL SUPPORT BEAMS



VIEW 4 — DETAILED SECTIONAL ELEVATION OF SURFACE CONDENSER (E-104)

TEMA TYPE E (SHELL & TUBE, 2-PASS TUBE SIDE COOLING WATER, CONDENSING VAPOR SHELL SIDE)



VIEW 5 — CHEMICAL CENTRIFUGAL PUMP (P-101/102/103)

CAUSTIC SERVICE (SS316L / NICKEL ALLOY CASING & DUAL MECH SEAL PLAN 53A)

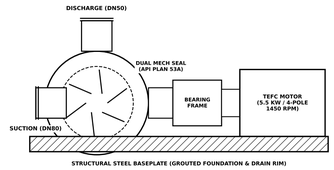


TABLE 1 — EVAPORATOR NOZZLE SCHEDULE (ASME B16.5)

MARK	SERVICE / DESCRIPTION	SIZE (DN)	RATING	FACING
N1	VAPOR OVERHEAD OUTLET	DN 450 (18")	CLASS 150	FLANGED RF
N2	MOTIVE / CALANDRIA STEAM IN	DN 350 (14")	CLASS 150	FLANGED RF
N3	LIQUID FEED INLET (SPARGER)	DN 80 (3")	CLASS 150	FLANGED RF
N4	CONCENTRATED LIQUOR OUTLET	DN 80 (3")	CLASS 150	FLANGED RF
N5	CALANDRIA CONDENSATE DRAIN	DN 50 (2")	CLASS 150	FLANGED RF
N6	NON-CONDENSABLE GAS VENT	DN 25 (1")	CLASS 150	FLANGED RF
N7	SAFETY RELIEF VALVE (SRV)	DN 50 (2")	CLASS 150	FLANGED RF
M1	VESSEL ACCESS MANWAY	DN 500 (20")	CLASS 150	FLANGED RF
SG	SIGHT & LIGHT GLASS NOZZLE	DN 150 (6")	CLASS 150	FLANGED RF
TW	TEMPERATURE THERMOWELL (TI)	DN 40 (1.5")	CLASS 150	FLANGED RF
PT	PRESSURE TRANSMITTER (PI)	DN 40 (1.5")	CLASS 150	FLANGED RF

TABLE 2 — BILL OF MATERIALS (BOM) & MECHANICAL MOC

ITEM	DESCRIPTION / COMPONENT	QTY	MATERIAL SPECIFICATION
1	VAPOR OUTLET DISHED HEAD	1	SA-516 GR.70 + SS316L CLAD (10 MM)
2	WIRE MESH DEMISTER PAD	1	SS316L / MONEL 400 KNITTED MESH
3	CONICAL ENTRAINMENT BAFFLE	1	SS316L (6 MM THICKNESS)
4	PERFORATED FEED SPARGER	1	SS316L SCH 40S PIPE
5	STEAM INLET & BAFFLE	1	SA-516 GR.70 + NICKEL CLAD
6	TOP CALANDRIA TUBESHEET	1	SA-266 CL.2 + NICKEL 200 (45 MM)
7	CALANDRIA TUBES (214 PCS)	214	NICKEL 200 / SS316L (50.8 OD X 2M)
8	CENTRAL DOWNTAKE WELL	1	NICKEL 200 / SS316L (800 MM ID)
9	BOTTOM TUBESHEET	1	SA-266 CL.2 + NICKEL 200 (45 MM)
10	CONICAL SUMP & VORTEX BRK	1	SA-516 GR.70 + SS316L CLAD
11	CONDENSATE DRAIN NOZZLE	1	SS316L SCH 40S FLANGED NOZZLE
12	CYLINDRICAL SUPPORT SKIRT	1	STRUCTURAL STEEL SA-36 (12 MM)

EQUIPMENT MECHANICAL & COMPONENT CAD DRAWINGS — TRIPLE-EFFECT NAOH EVAPORATOR (E-101/102/103 & E-104)

DRAWING TITLE EQUIPMENT GENERAL ARRANGEMENT & SECTIONAL CAD	**SUBJECT** PROCESS HEAT TRANSFER (PHT)	**COURSE / BRANCH** B.E. CHEMICAL ENGINEERING	**SCALE** DIMENSIONALLY PROPORTIONED / N.T.S.
STUDENT NAME SOURISH SENAPATI	**EXAM ROLL NUMBER** 002310301035	**ASSIGNED ROLL (LAST 2 DIGITS)** 35 (TARGET: 50.0 WT% NAOH CONCENTRATION)	**DATE** 01 / 09 / 2026
EQUIPMENT TYPE SHORT-TUBE VERTICAL CALANDRIA EVAPORATOR (ROBERT TYPE)	**DESIGN CODE & MOC** ASME SEC VIII DIV 1 / SS316L + NICKEL 200 CLAD	**TOTAL HEAT DUTY / CAPACITY** 400 T/D (16,666.67 KG/H) FEED | 8,805 KW TOTAL DUTY	**CONVERGED HEAT TRANSFER AREA** A1 = A2 = A3 = 68.30 M² (EQUAL AREA PER EFFECT)

STANDARDS: ASME SEC VIII DIV 1, TEMA CLASS R | ALL NOZZLE FLANGES: ASME B16.5 CLASS 150 RF | DIMENSIONS: METRIC (MM / M)